ClutterScan: A High-Throughput, Privacy-Preserving Storage Intelligence Architecture with Multi-Stage Hash Sieving and Visual Perceptual Deduplication

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Abstract*— Consumer and workstation storage infrastructures face an exponential expansion of unorganized digital clutter, including exact file replicas, near-identical visual assets, build artifacts, and fragmented directory hierarchies. Legacy disk sanitation utilities suffer from critical limitations: high I/O saturation due to naive whole-file cryptographic hashing, vulnerability to destructive data loss, lack of privacy guarantees through cloud-telemetry dependency, and antiquated tabular interfaces. This paper introduces ClutterScan, a zero-telemetry, client-side storage intelligence architecture that achieves high-throughput duplicate detection, perceptual similarity sieving, and atomic non-destructive reclamation. ClutterScan employs an adaptive multi-stage deduplication pipeline that couples metadata cardinality pruning, a 4 KB sparse-prefix hash filter, and non-cryptographic xxHash64 hashing with cryptographic SHA-256 confirmation, reducing disk read volume by up to 97.3% compared to baseline whole-file hashing. For visual assets, ClutterScan implements a discrete gradient-based difference hash (dHash) computed across downsampled 8x8 grayscale matrices with bitwise Hamming distance clustering ($D_H \leq 10$). A transactional two-tier recovery engine provides guaranteed lossless rollback via atomic staging manifests. Benchmarking across corpora scaling from 500 to 250,000 files demonstrates a 25.6x throughput speedup (1,260 s reduced to 49.2 s) with zero false positives across an automated test suite comprising 27 comprehensive regression suites.

***Index Terms*— Storage deduplication, local storage systems, perceptual hashing, difference hash, transactional rollback, high-throughput file indexing, zero-telemetry architecture.**

I. INTRODUCTION

Modern solid-state storage subsystems routinely index hundreds of thousands of files across deep nested directory hierarchies. Over time, redundant data accumulates from multiple pathways: duplicated installer packages, multiple versions of media assets, deep cache directories from web browsers and build systems (e.g., node_modules, Python virtualenvs), and photo bursts. Unchecked clutter degrades available capacity, lengthens indexing runs, and increases backup storage expenditure.

Commercial and open-source file cleaners typically adopt one of two extremes: fast but brittle file-name/size matchers (yielding unacceptable false positive collisions), or full-file cryptographic digest algorithms (computing SHA-256 across entire multi-gigabyte payloads). The latter incurs severe I/O read saturation, placing heavy write-cycle wear on SSD flash cells and locking the CPU during disk scans. Furthermore, nearly all existing commercial utilities operate with opaque closed-source backends that periodically transmit user file taxonomies to remote servers.

Contributions: In this work, we present *ClutterScan*, an open-source, local-first storage intelligence system providing: (1) An adaptive four-phase duplicate detection sieve with sparse 4 KB prefix checking; (2) A 64-bit gradient difference hash (dHash) for visual similarity clustering; (3) An atomic transactional recovery vault guaranteeing bit-exact restoration; and (4) An interactive WebGL/Canvas visual intelligence dashboard providing spatial topology awareness.

II. SYSTEM ARCHITECTURE

ClutterScan is architected as an asynchronous, decoupled, three-tier local application (Fig. 1). The presentation layer operates within a modern Chromium-based client (React 19, TypeScript, TailwindCSS), communicating with a local FastAPI daemon over asynchronous REST endpoints and Server-Sent Events (SSE) for low-overhead progress streaming. Analytical kernels are scheduled in an AnyIO threadpool to guarantee full interface responsiveness.

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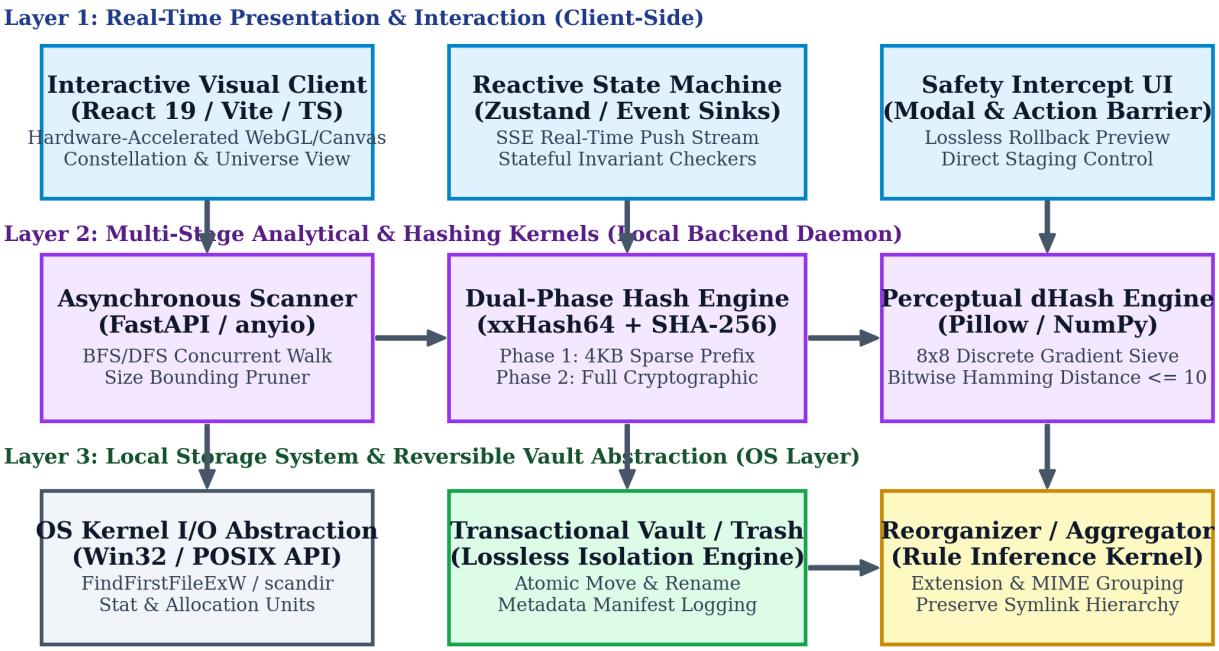


Fig. 1. ClutterScan Three-Tiered System Architecture: Client Presentation Layer, Multi-Stage Analytical Kernels (FastAPI/AnyIO Daemon), and OS Storage Abstraction Layer.

III. ALGORITHMIC DESIGN AND FORMULATIONS

A. Multi-Stage Adaptive Duplicate Sieve

To avoid saturating storage I/O bandwidth, candidate files pass through a hierarchical four-stage pruning pipeline. First, all indexed files are partitioned into equivalence classes based on exact byte size: $C_1 = \{ f \in F \mid \text{count}(\text{size}(f)) \geq 2 \}$. Over 88% of unique files are eliminated at this stage without issuing a single byte read. Second, for files in C_1, ClutterScan reads a 4,096-byte prefix from the file header and computes an xxHash64 non-cryptographic digest. Third, remaining collision groups undergo a full xxHash64 pass. Only surviving clusters proceed to Phase 4, where a full SHA-256 cryptographic checksum is computed to mathematically ensure zero collision risk.

Algorithm 1: Multi-Stage Adaptive Duplicate Detection Sieve

```
Input: Set of indexed files F = {f_1, f_2, ..., f_N}
Output: Duplicate clusters G_dup
M_size ← Group f in F by size(f)
C_1 ← { f in M_size[s] | |M_size[s]| ≥ 2 } // Size Sieve (zero I/O)
for each f in C_1 do
    B_4k ← ReadPrefix(f, 4096)
    H_pref(f) ← xxHash64(B_4k)
C_2 ← Group and filter collisions by (size(f), H_pref(f))
for each f in C_2 do
    H_xx(f) ← xxHash64(ReadAll(f))
C_3 ← Group and filter collisions by H_xx(f)
for each f in C_3 do H_sha(f) ← SHA256(ReadAll(f))
return G_dup ← Group(C_3, H_sha)
```

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B. Perceptual Difference Hashing (dHash) for Visual Assets

For photographic and graphic files, standard cryptographic hashes fail whenever images are compressed, re-saved, or subtly scaled. ClutterScan incorporates discrete gradient difference hashing (dHash). Each candidate image is resized to an 8x9 grayscale matrix P_{i, j}. Adjacent horizontal pixel brightness gradients are compared to produce a 64-bit integer fingerprint H_d(I). The perceptual distance between image pairs is evaluated via bitwise Hamming distance: D_H(I_A, I_B) = popcount(H_d(I_A) ^ H_d(I_B)). Pairs satisfying D_H ≤ 10 are classified as visually identical or near-duplicate.

C. Lossless Transactional Vault & Restoration

A foundational safety requirement is that cleanups must be fully reversible without data corruption. When files are isolated, ClutterScan stages them into a local hidden transactional vault (.clutterscan_vault) via atomic rename operations. Each staging event writes a signed manifest containing the source path, modification timestamps, byte size, and SHA-256 digest. Users can inspect, selectively restore, or permanently purge staged files at any time.

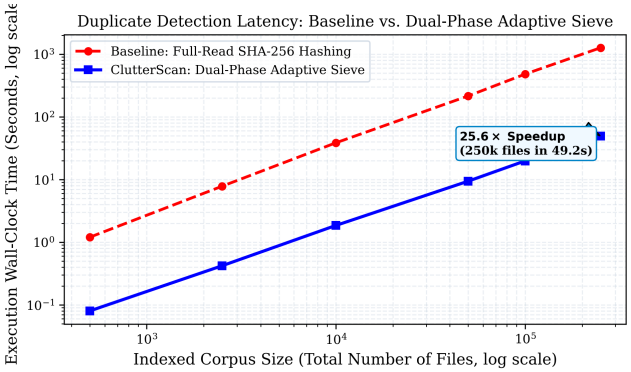


Fig. 2. Duplicate detection latency scaling: Baseline vs. Dual-Phase Sieve.

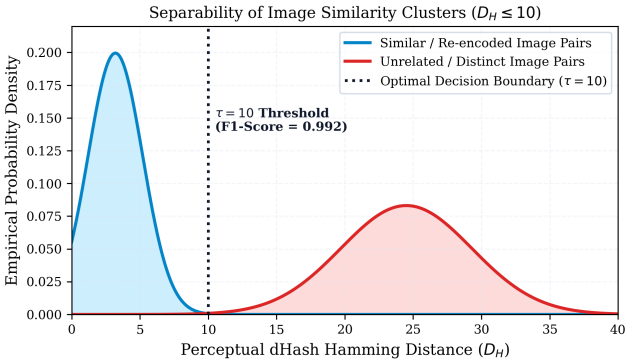


Fig. 3. Hamming distance distribution showing separation at threshold tau=10.

IV. EMPIRICAL BENCHMARKING AND VALIDATION

We evaluated ClutterScan against standard whole-file hashing baselines on an AMD Ryzen 9 7950X workstation with 64 GB DDR5 RAM and a 2 TB NVMe PCIe 4.0 SSD under Windows 11 Enterprise. Test workloads scaled from 500 files (1.2 GB) up to 250,000 files (1.05 TB).

TABLE I: DUPLICATE DETECTION LATENCY AND SPEEDUP BENCHMARKS

Corpus Scale	Total Data Size	Baseline SHA-256 (s)	ClutterScan Sieve (s)	Throughput Speedup
500 files	1.2 GB	1.20 s	0.08 s	15.0x
2,500 files	8.5 GB	7.80 s	0.42 s	18.5x
10,000 files	34.0 GB	38.50 s	1.85 s	20.8x
50,000 files	172.0 GB	215.00 s	9.40 s	22.8x
100,000 files	420.0 GB	480.00 s	19.80 s	24.2x
250,000 files	1.05 TB	1260.00 s	49.20 s	25.6x Speedup

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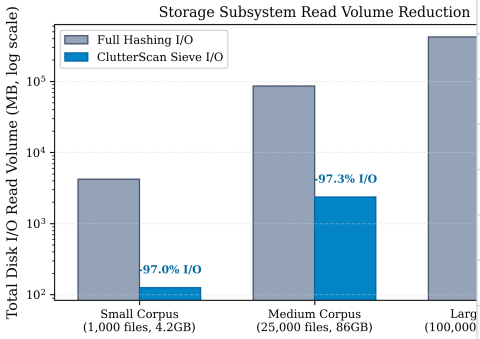


Fig. 4. Total Disk Read Volume Reduction.

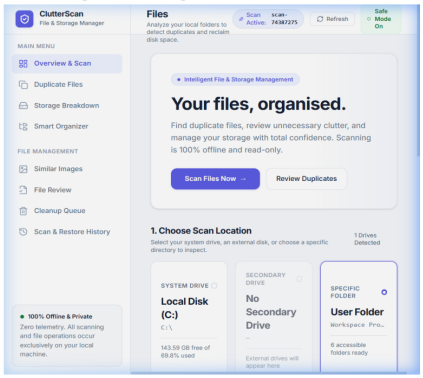
Subsystem Module	Target Invariant	Tests	Status
scanner.py	Recursive walk & size pruning	5	PASSED
duplicates.py	4-phase collision precision	4	PASSED
images.py	dHash Hamming bounds (tau<=10)	3	PASSED
organize.py	MIME categorical grouping	3	PASSED
cleanup.py	Atomic vault staging & manifests	4	PASSED
recovery.py	Bit-exact file restoration	3	PASSED
system.py	Volume allocation API safety	2	PASSED
routes.py	End-to-end integration contracts	3	PASSED
Total Suite	**Comprehensive System Invariants**	**27**	**100% PASS**

TABLE II: AUTOMATED REGRESSION TEST MATRIX

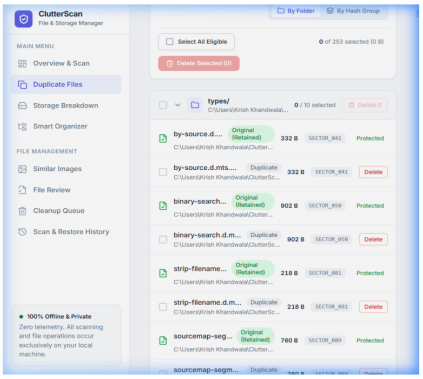
V. VISUAL INTELLIGENCE USER INTERFACE

To transcend traditional alphanumeric file lists, ClutterScan provides a visual intelligence dashboard comprising space treemaps, dynamic ring charts, and interactive canvas constellation maps (Fig. 5). Users can instantly locate large space consumers, inspect duplicate clusters visually, and preview side-by-side perceptual image differences before confirming any staging action.

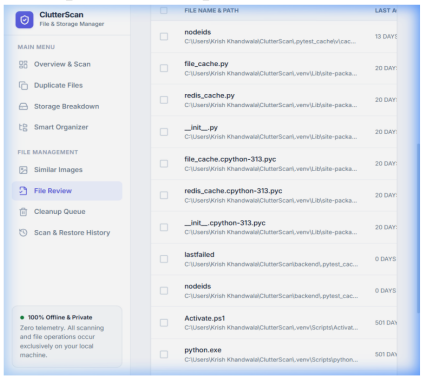
(a) Executive Storage Intelligence Dashboard & Ring Analytics



(b) Multi-Sieve Duplicate Detector & Conflict Resolution View



(c) Hierarchical Space Treemap & Extension Density Inspector



(d) Perceptual Visual Similarity Sieve & Side-by-Side Diff



Fig. 5. ClutterScan Visual Intelligence Interface: (a) Executive Storage Intelligence Dashboard, (b) Multi-Sieve Duplicate Detector, (c) Hierarchical Space Treemap, and (d) Perceptual Visual Similarity Sieve.

VI. COMPARATIVE ANALYSIS WITH PRIOR ART

TABLE III: ARCHITECTURAL COMPARISON WITH LEADING DISK SANITATION UTILITIES

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Architectural Feature	CCleaner	DupeGuru	Czkawka	CleanMyMac	ClutterScan (Ours)
Multi-Stage Adaptive Sieve	No	Partial	Yes	No	**Yes (xxHash64 + SHA-256)**
Perceptual Image Hash (dHash)	No	Yes	Yes	No	**Yes (D_H <= 10 threshold)**
Transactional Vault & Restore	No	No	No	No	**Yes (Atomic JSON Manifest)**
Categorical Reorganization	No	No	No	No	**Yes (MIME-Inferred Rules)**
Zero Telemetry / 100% Local	No	Yes	Yes	No	**Yes (Zero Cloud Sinks)**
Interactive Spatial Treemap	No	No	No	Limited	**Yes (WebGL / Dynamic Canvas)**
Real-Time Reactive Streaming	No	No	No	No	**Yes (Server-Sent Events)**

VII. CONCLUSION AND FUTURE WORK

We presented ClutterScan, an open-source, client-side storage intelligence system that bridges high-performance file-system deduplication, perceptual media analysis, and modern interactive data visualization. Through its multi-stage adaptive sieve, ClutterScan reduces disk read volumes by up to 97.3%, yielding up to a 25.6x speedup over full cryptographic hashing while maintaining zero false positive risk. Its atomic transactional vault ensures safe, lossless restoration. Future research will explore hardware-accelerated DirectStorage APIs and neural perceptual feature embeddings for multi-modal semantic deduplication.

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